

designed for artificial intelligence applications. Qiskit, TensorFlow Quantum, and PennyLane are a few instances of frameworks that aid in the formulation and optimisation of algorithms.

Quantum data structures refer to algorithms and structures that have been specifically engineered to efficiently represent and manipulate quantum data. These frameworks facilitate the manipulation, retrieval, and storage of quantum data, which is of the utmost importance for tasks involving quantum machine learning and pattern recognition.

Notwithstanding its potential, quantum AI encounters a number of obstacles that impede its extensive implementation and scalability.

In order to guarantee the dependability and precision of computations, robust error correction techniques and fault-tolerant quantum hardware are required for quantum systems, as these are susceptible to noise, decoherence, and errors.

One common limitation of quantum processors is their restricted qubit connectivity, which imposes a hindrance on the execution of intricate quantum circuits and algorithms. To overcome limitations imposed by connectivity, it is imperative to devise inventive qubit architectures and optimise circuits.

Current applications of quantum AI

Quantum AI, despite being in its nascent stages, has exhibited encouraging implementations in a multitude of domains (see Table 1).

Drug discovery. Drug discovery is expedited through the utilisation of quantum algorithms, which optimise molecular structures, identify potential drug candidates, and predict molecular properties with extreme precision.

Financial modelling. By facilitating option pricing, portfolio

Industry	Quantum AI Applications
Healthcare	Drug discovery: Accelerating molecular simulations for drug discovery, optimising drug design and discovery processes.
	Medical imaging: Enhancing image processing and analysis techniques for medical imaging, improving diagnostic accuracy and treatment planning.
	Genomics: Analysing genomic data and identifying genetic markers associated with diseases, personalised medicine and precision healthcare.
Finance	Risk assessment: Faster risk assessment and portfolio optimisation in financial markets, improving decision-making processes and reducing financial risks.
	Algorithmic trading: Developing quantum algorithms for high-frequency trading, optimising trading strategies and enhancing market prediction capabilities.
	Portfolio management: Quantum-enhanced portfolio optimisation techniques for asset allocation and risk management, maximising investment returns.
Logistics	Supply chain optimisation: Quantum algorithms for route optimisation, inventory management, and supply chain logistics, reducing operational costs and improving efficiency.
	Demand forecasting: Predicting demand patterns and optimising inventory levels using quantum machine learning techniques, minimising stockouts and overstocking.
	Warehouse management: Quantum-inspired algorithms for warehouse layout optimisation and inventory management, enhancing storage efficiency and order fulfilment.
Energy	Energy distribution: Optimising energy distribution networks and grid management using quantum algorithms, facilitating the integration of renewable energy sources.
	Energy trading: Developing quantum algorithms for energy trading and pricing, optimising energy markets and promoting sustainable energy systems.
	Smart grids: Implementing quantum-inspired solutions for smart grid optimisation and demand response management, enhancing grid reliability and efficiency.
Manufacturing	Process optimisation: Quantum algorithms for process optimisation and predictive maintenance in manufacturing, improving production efficiency and reducing downtime.
	Quality control: Quantum-enhanced quality control techniques for defect detection and product inspection, ensuring product quality and reliability.
	Supply chain visibility: Leveraging quantum algorithms for supply chain visibility and traceability, enhancing transparency and accountability in manufacturing processes.
Telecommunications	Network optimisation: Quantum-inspired algorithms for network optimisation, routing, and resource allocation in telecommunications networks, improving network performance.
	Spectrum management: Quantum algorithms for spectrum management and interference mitigation in wireless communication systems, optimising spectrum utilisation.
	Network security: Enhancing network security through quantum-resistant encryption and quantum-inspired security protocols, safeguarding communication channels.

optimisation, and risk assessment in financial markets at a quicker rate, quantum algorithms improve decision-making processes and mitigate financial risks.

Cybersecurity. Quantum-enhanced cryptography provides secure communication protocols resistant to quantum attacks, assuring the confidentiality, integrity, and authenticity of data transmission.

Energy optimisation. By optimising energy distribution networks,

resource allocation, and grid management, quantum algorithms reduce carbon emissions and facilitate the transition to sustainable energy systems.

Challenges in developing quantum AI algorithms

Quantum decoherence. Imagine building a house of cards in a hurricane. Quantum systems are just as delicate. Even the slightest environmental noise can disrupt the fragile